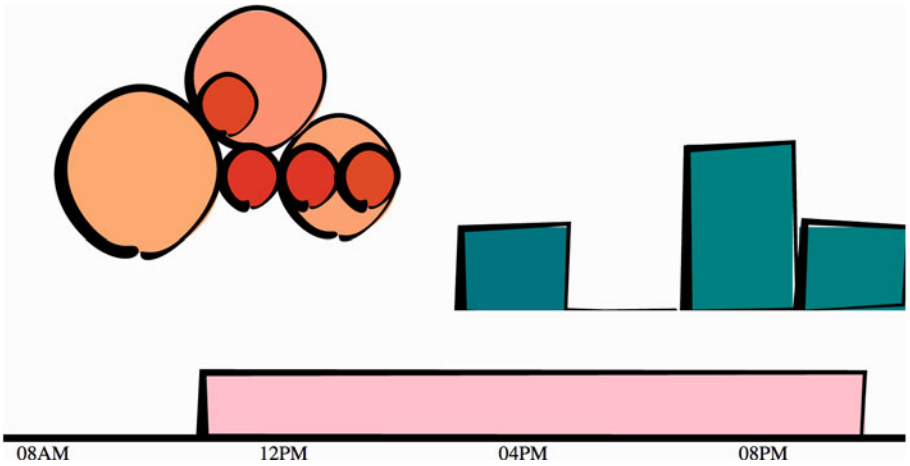


Figure 1.8 You can combine these layouts and functions to create a data dashboard like we'll do in chapter 9. You can also use the drawing functions to make your bar charts look distinctive, such as this "sketchy" style.

Although the ability to create rich and varied graphics is one of D3's strong points, more important for modern web development is the ability to embed the high level of interactivity that users expect. With D3, every element of every chart, from a spinning globe to a single, thin slice of a pie chart, is made interactive in the same way. And because D3 was written by someone well versed in data visualization practice, it includes a number of interactive components and behaviors that are standard in data visualization and web development.

You don't invest your time learning D3 so that you can deploy Excel-style charts on the web. For that, there are easier, more convenient libraries. You learn D3 because it gives you the ability to implement almost every major data visualization technique. It also gives you the power to create *your own* data visualization techniques, something a more general library can't do.

For more examples of the variety of different data visualization techniques realized with D3, take a look at Christophe Viau's gallery of over 2,000 D3 examples here: http://christopheviau.com/d3list/gallery.html.

By requiring a break with the practice of supporting long-obsolete browsers, D3.js affords developers the capacity to make not only richly interactive applications but also applications that are styled and served like traditional web content. This makes them more portable, more amenable to the growing, linked data web, and more easily maintained by large teams.

The decision on Bostock's part to deal broadly with data, and to create a library capable of presenting maps as easily as charts, as easily as networks, as easily as ordered lists, also means that a developer doesn't need to try to understand the abstractions and syntax of one library for maps, and another for dynamic text content, and another for data visualization. Instead, the code for running an interactive, force-directed network